Nilai UNIVERSITY

BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION
FEBRUARY 2025

Course : EC3105 (C Programming)	Time: 8.00 am – 11.00 pm (3 hours)
Lecturer : Dr. Nur Inshirah Naqiah	Date : 26th May 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 5 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. Define the following system calls used in Unix programming and explain their primary purpose: **(6 MARKS)**

   i. 'execvp()'

   ii. 'pipe()'

   iii. 'getpid()'

   b. Explain the purpose of the systems mentioned in (a). **(3 MARKS)**

   c. Define the following Unix system calls used for file operations: **(8 MARKS)**

   i. 'open()'

   ii. 'read()'

   iii. 'write()'

   iv. 'close()'

   d. Explain the purpose of the systems mentioned in (c). **(8 MARKS)**

2. a. Define the following arithmetic operators used in C programming: **(5 MARKS)**

   i. '+' (Addition)

   ii. '-' (Subtraction)

   iii. '*'(Multiplication)

   iv. '/' (Division)

   v. '%' (Modulus)

   b. Write a C program that demonstrates the use of selection constructs (i.e., if, if-else, and switch). The program should: **(10 MARKS)**

   - Prompt the user to enter a number.
   - Check if the number is positive, negative, or zero using

an if-else construct and print the appropriate message.

- Check if the number is even or odd using an if-else construct and print the appropriate message.
- Use a switch statement to categorize the number into "small" (less than 10), "medium" (between 10 and 100), or "large" (greater than 100) and print the corresponding category.

c. Write a C program that uses selection constructs to determine the grade based on the score provided by the user. The program should: **(5 MARKS)**

- Prompt the user to enter a score (an integer between 0 and 100).
- Use an if-else construct to determine the grade according to the following criteria:
    - o Score >= 90: Grade A
    - o Score >= 80 and < 90: Grade B
    - o Score >= 70 and < 80: Grade C
    - o Score >= 60 and < 70: Grade D
    - o Score < 60: Grade F
- Print the corresponding grade.

d. Write a C program that prompts the user to enter an integer representing a month (1 to 12) and then uses a switch statement to print the name of the month. Additionally, handle invalid input (i.e., numbers outside the range 1 to 12) by printing an error message. **(5 MARKS)**

3. a. Write a C program that uses iteration constructs to perform the following tasks: **(8 MARKS)**

- Prompt the user to enter a positive integer ‘n’.
- Print all integers from ‘1’ to ‘n’ (inclusive) using a for loop.
- Calculate and print the sum of all integers from ‘1’ to ‘n’ using a while loop.
- Print all even integers from ‘1’ to ‘n’ using a do-while loop.

b. Write a C program that defines and uses functions to perform the following tasks: **(6 MARKS)**

- Define a function 'int power(int base, int exponent)' that calculates and returns the value of 'base' raised to the power of exponent.
- Define a function 'float average(int a, int b, int c)' that calculates and returns the average of three integers.
- In the 'main()' function, prompt the user to enter two integers for base and exponent, and then calculate and print the result of raising the base to the power of the exponent using the 'power()' function.
- Prompt the user to enter three integers and then calculate and print their average using the 'average()' function.

c. Write a C program that includes functions to perform the following tasks: **(6 MARKS)**

- Define a function 'int square(int x)' that returns the square of an integer.
- Define a function 'void print_hello()' that prints "Hello, World!" to the console.
- In the 'main()' function, prompt the user to enter an integer, use the 'square()' function to compute and print the square of that integer.
- Call the 'print_hello()' function to print "Hello, World!" after printing the square of the integer.

d. Write a C program that includes functions to perform the following tasks: **(5 MARKS)**

- Define a function 'float calculate_area(float radius)' that calculates and returns the area of a circle given its radius. Use the formula area=π×radius2 (where π≈3.14).
- Define a function 'void print_message()' that prints "Calculation complete!" to the console.
- In the 'main()' function, prompt the user to enter the radius of a circle. Use the 'calculate_area()' function to compute and print the area of the circle. After that, call the 'print_message()' function to print the message.

4. a. Write a C program that performs the following tasks: **(10 MARKS)**

- Define a structure 'Student' to store the following information:
  - 'name' (a string of up to 50 characters)
  - 'roll_no' (an integer)
  - 'marks' (an array of 5 floating-point numbers representing the marks in 5 subjects)
- Define a function 'void input_student_data(struct Student*s)' that takes a pointer to a 'Student' structure as a parameter and allows the user to input the student's name, roll number, and marks for 5 subjects.
- Define a function 'float calculate_average(struct Student*s)' that takes a 'Student' structure as a parameter and returns the average of the 5 subject marks.
- In the 'main()' function:
  - Create an array of 3 'Student' structures.
  - Use a loop to call 'input_student_data()' for each student to input their data.
  - Use another loop to calculate and print the average marks for each student using the 'calculate_average()' function.

b. Write a C program that performs the following tasks: **(9 MARKS)**

- Define a function 'void string_copy(char*dest, const char*src)' that copies the contents of one string (source) to another (destination).
- Define a function 'int string_length(const char*str)' that returns the length of a string.
- In the 'main()' function:
  - Prompt the user to enter a string.
  - Use the 'string_length()' function to compute and print the length of the string.
  - Use the 'string_copy()' function to copy the string to another string.
  - Print the copied string.

c. Write a C program that performs the following file operations: **(6 MARKS)**

- Define a function 'void write_numbers_to_file(const char *filename, int num1, int num2)' that writes two integers to a file, each on a new line.
- Define a function 'void read_numbers_from_file(const char *filename)' that reads the two integers from the file and prints them to the console.

- In the 'main()' function:
    - Call 'write_numbers_to_file()' to write two integers (e.g., 15 and 30) to a file named 'numbers.txt'.
    - Call 'read_numbers_from_file()' to read and print the integers from 'numbers.txt'.

-END OF QUESTION PAPER-